Research Plan: Quantum Reinforcement Learning for Decision Making

QIntern 2025 Mentor Application

Overview

This project focuses on building and analyzing a simple Quantum Reinforcement Learning (Q-RL) agent for decision-making tasks. Reinforcement learning trains an agent to take actions in an environment and improve its decisions using rewards. In this project, a variational quantum circuit will be used as a policy model or value-function component and compared with classical reinforcement learning baselines.

The goal is to implement a small, realistic prototype using simulation tools and study whether a quantum policy model can learn useful behavior on simple benchmark environments such as CartPole or GridWorld. The project will focus on implementation, benchmarking, and clear analysis rather than claiming quantum advantage.

Project Title

Q-RL Agent: Quantum Reinforcement Learning for Decision Making

Duration

6 Weeks

Intern Level

Early professional career with basic Python, AIML, reinforcement learning, and introductory quantum computing background.

Objective

To design, implement, and evaluate a hybrid quantum-classical reinforcement learning agent for a simple decision-making environment. The intern will compare the Q-RL model with a classical RL baseline and analyze learning performance using reward curves, convergence behavior, and model limitations.

Six-Week Work Plan

Week 1: Background and Environment Setup

- Study reinforcement learning basics: agent, environment, state, action, reward, policy, and value function.
- Review basic quantum computing concepts: qubits, gates, measurement, variational circuits, and parameterized quantum circuits.
- Install Python tools such as Gymnasium, NumPy, Matplotlib, Qiskit or PennyLane.
- **Deliverable:** Short summary of RL and Q-RL concepts with completed setup.

Week 2: Classical RL Baseline

- Select a simple environment such as CartPole-v1 or a custom GridWorld.
- Implement or adapt a classical RL baseline such as policy gradient, DQN, or Q-learning.
- Run baseline experiments and record reward curves and success rate.
- **Deliverable:** Working classical RL baseline with initial plots.

Week 3: Quantum Policy Model Design

- Design a small variational quantum circuit for representing the policy or action-selection module.
- Encode environment states into quantum circuit parameters using angle encoding.
- Connect quantum circuit outputs to action probabilities using a classical optimizer.
- **Deliverable:** Basic Q-RL model architecture and circuit diagram.

Six-Week Work Plan Continued

Week 4: Q-RL Implementation and Training

- Train the hybrid Q-RL agent in the selected environment.
- Tune basic parameters such as learning rate, circuit depth, number of qubits, and number of episodes.
- Compare the learning behavior of classical RL and Q-RL models.
- **Deliverable:** Training results for Q-RL agent and comparison with baseline.

Week 5: Evaluation and Analysis

- Evaluate models using average reward, convergence speed, stability, and success rate.
- Study the effect of circuit depth and number of qubits on performance.
- Discuss practical limitations such as simulation cost, noise, and scalability.
- **Deliverable:** Comparative result tables, plots, and observations.

Week 6: Documentation and Final Report

- Prepare final report explaining methodology, experiments, results, and limitations.
- Organize the code with clear instructions for running experiments.
- Prepare a short presentation summarizing the project and future research scope.
- **Deliverable:** Final report, code repository, graphs, and presentation.

Guidance and Supervision Plan

Weekly meetings will be conducted to review progress, debug implementation issues, and discuss experimental results. Guidance will be provided on reinforcement learning basics, quantum circuit design, result interpretation, and scientific writing. Additional feedback sessions may be scheduled during the implementation and final-report stages.

Tools and Resources

- Python, NumPy, Pandas, Matplotlib
- Gymnasium or OpenAI Gym for reinforcement learning environments
- Qiskit or PennyLane for quantum circuit simulation
- PyTorch or TensorFlow for optional neural network baselines
- Reference topics: policy gradient methods, variational quantum circuits, hybrid quantum-classical learning

Expected Outcomes

- A working prototype of a Q-RL agent for a simple decision-making task.
- A baseline comparison between classical RL and quantum-enhanced RL.
- Training graphs, reward curves, and performance tables suitable for a preliminary research paper.
- A clear discussion of where Q-RL is promising and where it is currently limited.

Plans for Future Work

- Extend the work to more complex environments and multi-agent decision-making tasks.
- Test Q-RL models under simulated quantum noise.
- Explore deeper variational circuits and different encoding strategies.
- Develop a paper based on the experimental results and comparison study.